Verilog HDL code for police siren

```
module police_siren (
    input clk,
    output reg [3:0] red_led,
    output reg [3:0] blue_led,
    output reg [1:0] white_led,
    output wire buzzer
);

    reg [23:0] led_counter = 0;    //for LED blinking
    reg [15:0] buzzer_counter = 0; //for buzzer frequency control
    reg siren_toggle = 0;          // for switching LED colors
    reg buzzer_toggle = 0;         //for buzzer frequency variation

    // clk divider for LED blinking
    always @(posedge clk)
            begin
        if (led_counter == 25000000) begin
            led_counter <= 0;
            siren_toggle <= ~siren_toggle;
        end
                    else begin
            led_counter <= led_counter + 1;
        end
    end

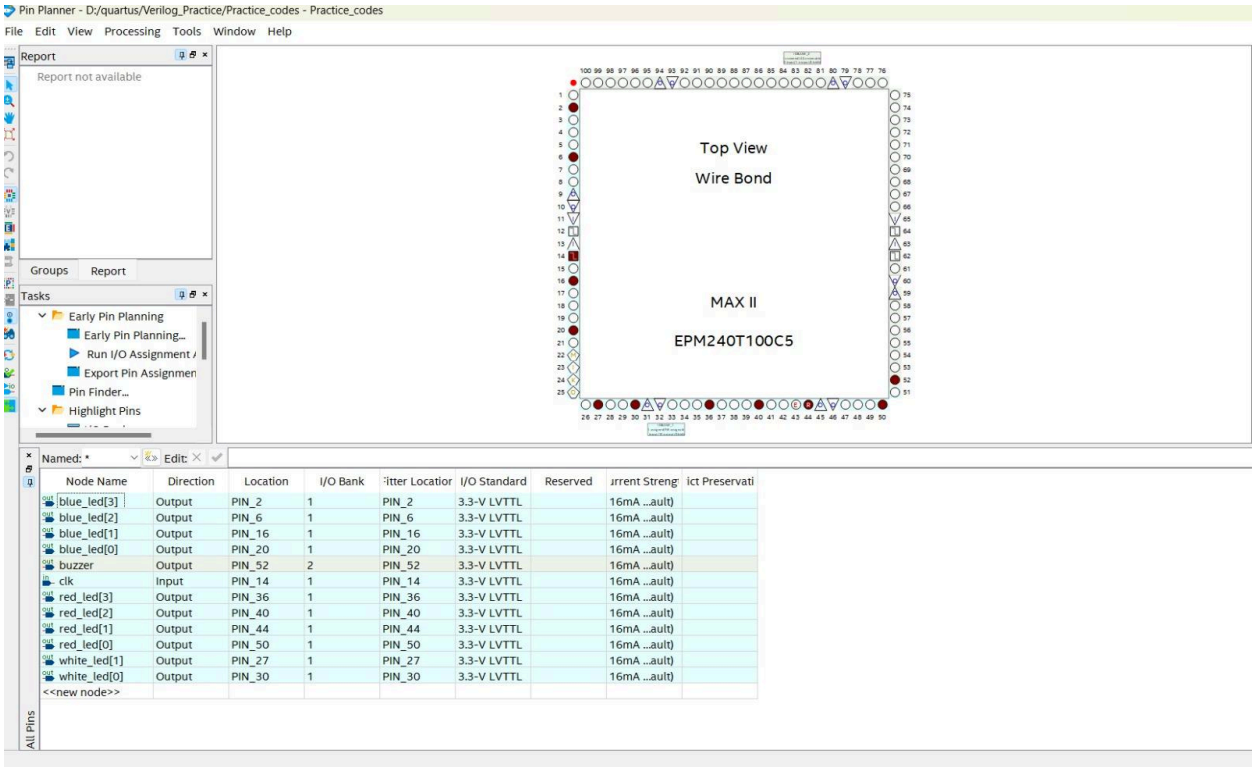
    // LED toggle
    always @(posedge clk)
            begin
        if (siren_toggle) begin
            red_led <= 4'b1111;   //all red LEDs on
            blue_led <= 4'b0000;  // all blue LEDs off
        end
                    else begin
            red_led <= 4'b0000;   // all red LEDs off
            blue_led <= 4'b1111;  //all blue LEDs on
```

```
        end
        white_led <= 2'b11;       // White LEDs always on
    end

    // Variable frequency buzzer for siren
    always @(posedge clk)
            begin
        if (buzzer_counter == 50000) begin
            buzzer_counter <= 0;
            buzzer_toggle <= ~buzzer_toggle;
        end
                    else begin
            buzzer_counter <= buzzer_counter + 1;
        end
    end

    assign buzzer = buzzer_toggle;

endmodule
```



Node Name	Direction	Location	I/O Bank	itter Locatior	I/O Standard	Reserved	urrent Streng	ict Preservati
blue_led[3]	Output	PIN_2	1	PIN_2	3.3-V LVTTL		16mA ...ault)	
blue_led[2]	Output	PIN_6	1	PIN_6	3.3-V LVTTL		16mA ...ault)	
blue_led[1]	Output	PIN_16	1	PIN_16	3.3-V LVTTL		16mA ...ault)	
blue_led[0]	Output	PIN_20	1	PIN_20	3.3-V LVTTL		16mA ...ault)	
buzzer	Output	PIN_52	2	PIN_52	3.3-V LVTTL		16mA ...ault)	
clk	Input	PIN_14	1	PIN_14	3.3-V LVTTL		16mA ...ault)	
red_led[3]	Output	PIN_36	1	PIN_36	3.3-V LVTTL		16mA ...ault)	
red_led[2]	Output	PIN_40	1	PIN_40	3.3-V LVTTL		16mA ...ault)	
red_led[1]	Output	PIN_44	1	PIN_44	3.3-V LVTTL		16mA ...ault)	
red_led[0]	Output	PIN_50	1	PIN_50	3.3-V LVTTL		16mA ...ault)	
white_led[1]	Output	PIN_27	1	PIN_27	3.3-V LVTTL		16mA ...ault)	
white_led[0]	Output	PIN_30	1	PIN_30	3.3-V LVTTL		16mA ...ault)	
<<new node>>								